

CSCI 4125/5125 Course Project
Data Models and Database Systems
Fall 2024
Course Project
Phase 10: Normalization

Due: Tuesday, 12/3 @ 11:59pm

Reading: Silberschatz Chapter 7

Submission Guidelines:

1. This assignment is worth 30 points for all students.
2. It is your responsibility to make sure all files are readable and submitted on time.

Submission:

- Tasks 1 & 2. Submit a single file for all answers. You may draw out solutions by hand, but they must be legible and clearly organized into a single file.

Task 1. Agent Relation (18 points). Since you originally designed the real estate database, fields have been added to the original Agent relation. You start to notice some update anomalies caused by data redundancy. It's up to you to evaluate the relation and fix any problems.

Original Agent Relation

AGENT(ID, Name, Phone, AssistantID)

New Agent Relation

AGENT(ID, Name, Phone, AssistantID, OfficeAddress, Neighborhood, OfficePhone, Manager, ParkingSpot, RoomNumber, Floor, View, Windows)

F = {
 AgentID, OfficeAddress → ParkingSpot
 AgentID → Name, Phone, AssistantID, RoomNumber
 OfficeAddress → Neighborhood, OfficePhone, Manager
 RoomNumber → Floor, View, Windows
}

1. What is the primary key?
2. Remove any partial key dependencies to create a set of linked relational schemas in Second Normal Form. Primary keys require a solid underline. Foreign keys require a dotted underline and an arrow to the attribute(s) they reference.
3. Remove any transitive dependencies to create a set of linked relational schemas in Third Normal Form. Primary keys require a solid underline. Foreign keys require a dotted underline and an arrow to the attribute(s) they reference.

Task 2. Seller Relation (12 points). You want to be able to store multiple phone numbers for each seller. As of now, you have one column to store a phone number in the seller table.

Original Seller Relation

SELLER(SSN, Name, Phone)

You may want to read over all questions before answering. These questions build on each other. Only provide the answer that applies to each question.

4. To allow multiple phone numbers for each seller, how would you alter the relation so that it is in proper 1NF? Show the proper 1NF schema. *Hint:* think back to the pizza toppings example.

5. Did your new relation in 1NF introduce any redundancies? If so, what functional dependencies do you need to decompose on? *Hint:* it may help to create some example values.

6. Using the functional dependencies you identified, show the new relation(s) in proper 3NF. Primary keys require a solid underline. Foreign keys require a dotted underline and an arrow to the attribute(s) they reference.